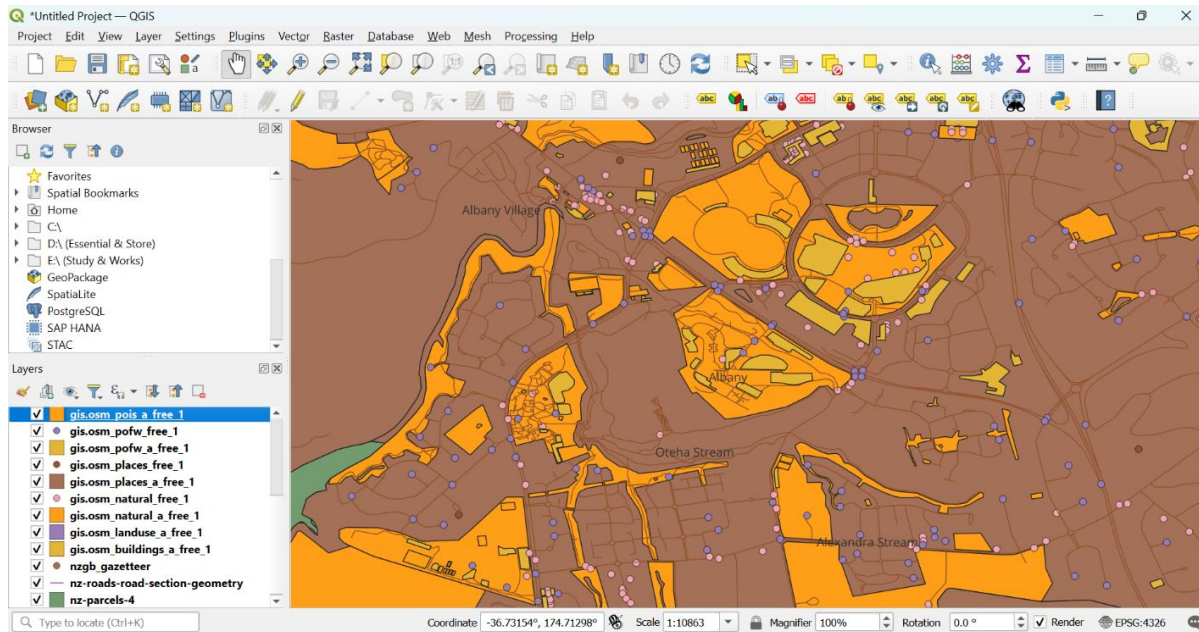
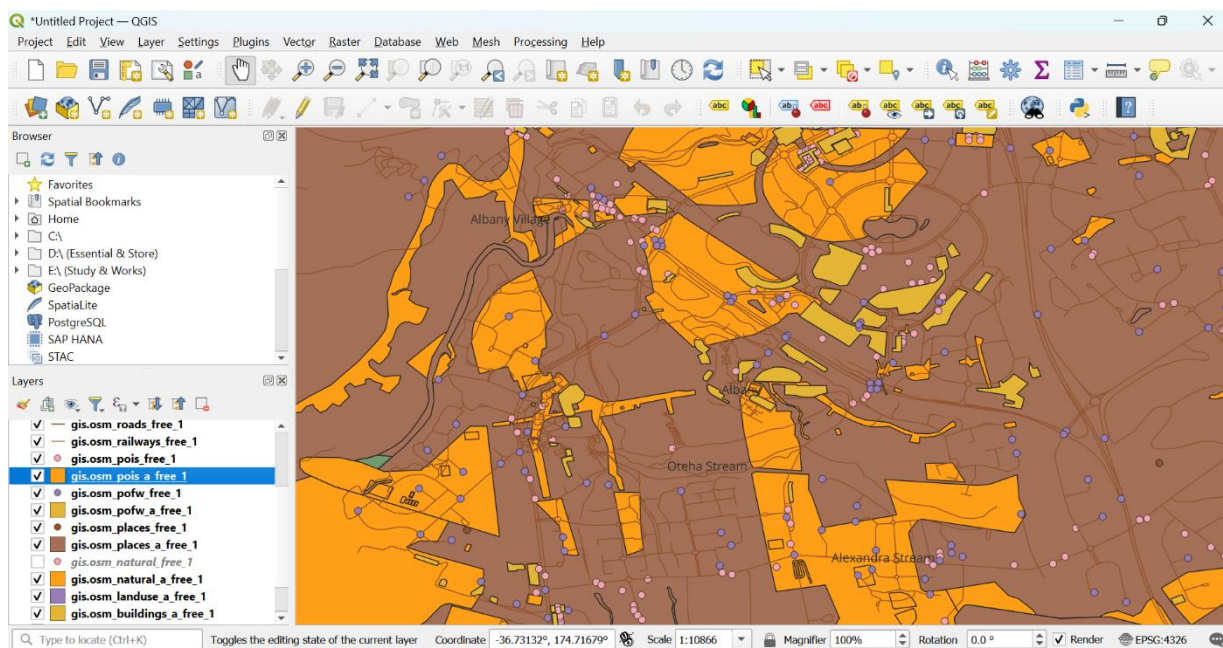


At-Home Activity 2: Map Projections

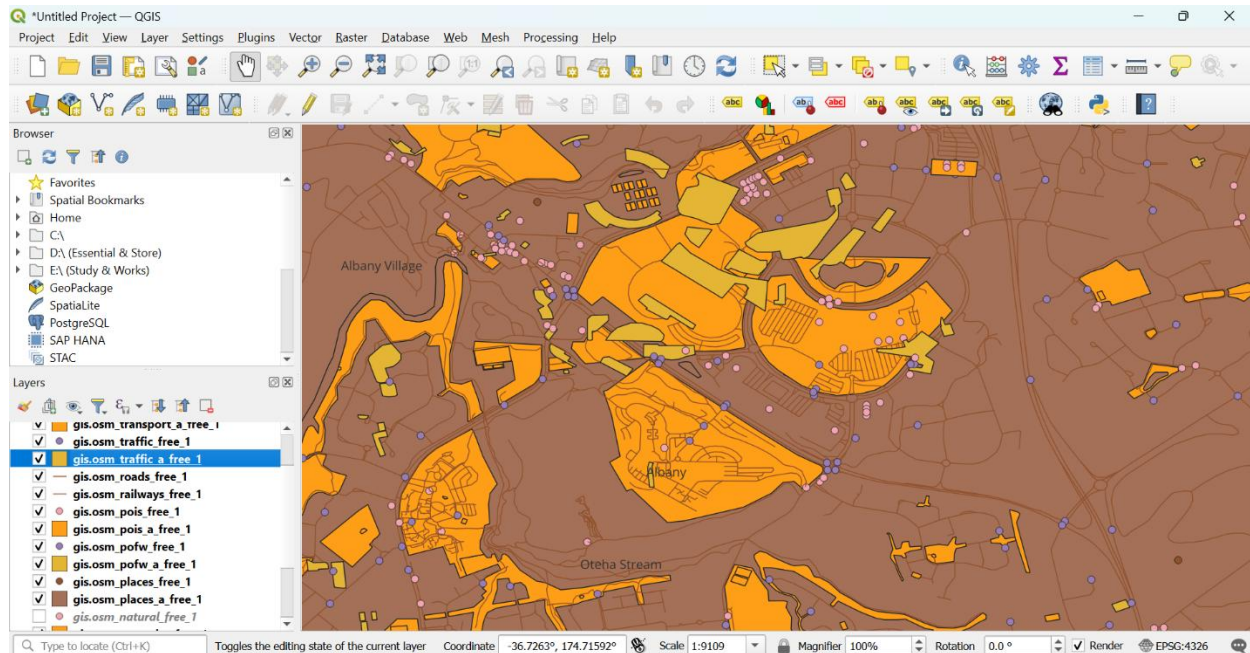
Started a new project at QGIS and added datasets **Open Street Map NZ**, **NZ parcels** (nz-parcels-4.shp) and **roads shape file**. Then added **place names** (Delimited Text Layer) file to the project. The map is now projected then zoomed into the Albany area where Massey University is located. The layers are aligned as expected and place names are shown as well. Here is the map view:



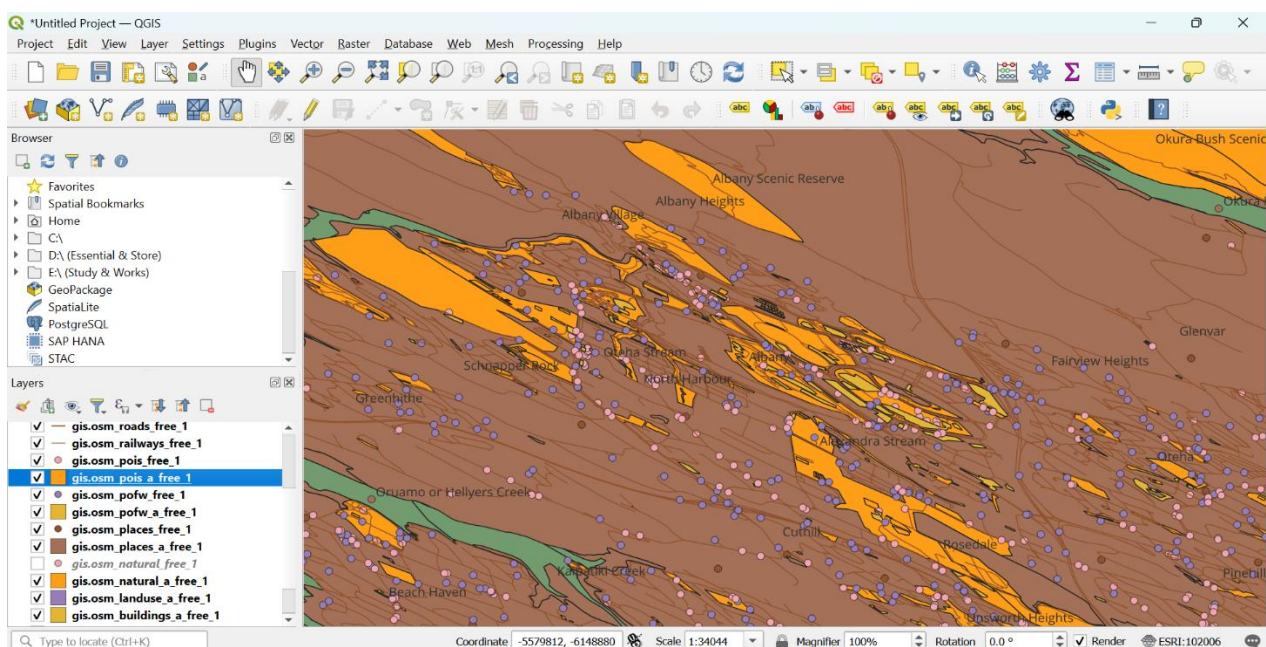
- a. I changed the map projection for the layer “gis.osm_pois_a_free_1” from EPSG:4326 to **EPSG:4310 - Yoff**. Then the alignment of the layer on map was moved to side and became incorrect. The misalignment happened because of the coordinate systems. EPSG:4326 is the standard global latitude-longitude system based on Lat/long (Geodetic alias) method. Though the EPSG:4310 is also based on the same method but it is static and relies on a datum which is plate-fixed. That’s why the layer projection on map was misaligned.



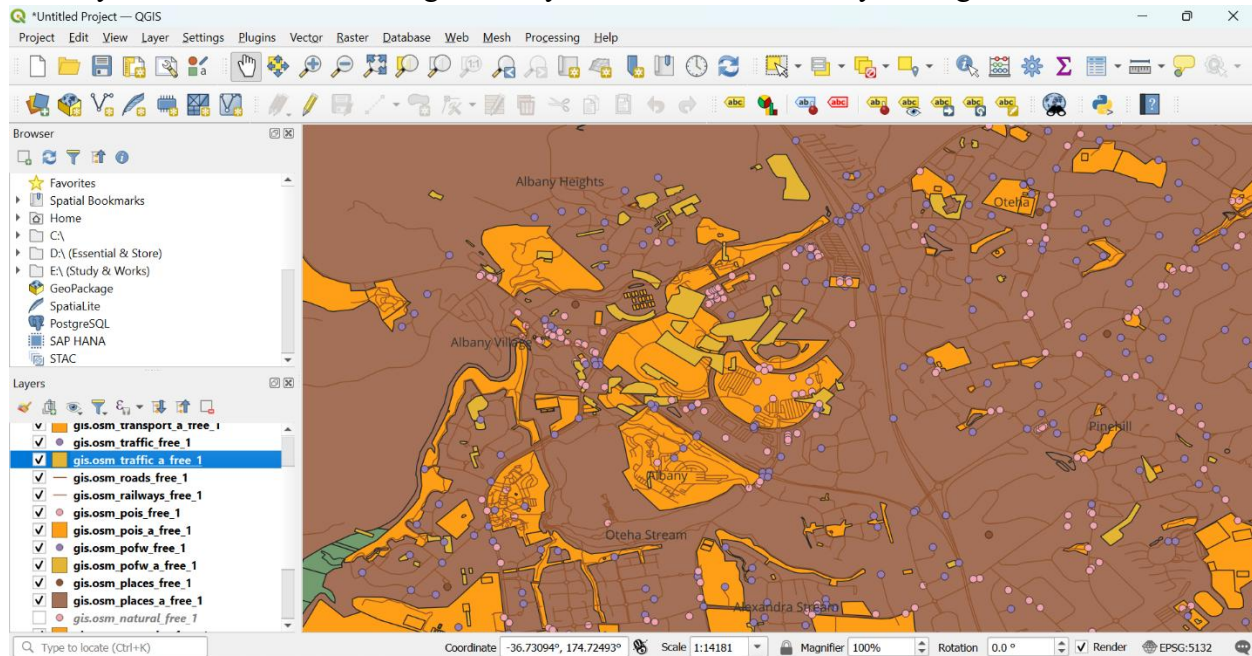
Then I changed the map projection to **EPSG:5132 Tokyo 1892** from EPSG:4326 for the layer “gis.osm_traffic_a_free_1”. Then again the alignment of the layer on map was moved to side and became incorrect. The misalignment happened because of the coordinate systems. EPSG:4326 is the standard global latitude-longitude system based on Lat/long (Geodetic alias) method. Though the EPSG:5132 is also based on the same method but it is static and relies on a datum which is plate-fixed. That’s why the layer projection on map was misaligned.



- b. I changed the projection for the entire map view this time. I changed the Coordinate Reference Systems (CRS) from the default EPSG:4326 to **ESRI:102006** from Alaska. Then the map was distorted and stretched. This static CRS is based on Albers Equal Area method. The projection is not now well understandable. This is because ESRI:102006 is optimised for Alaska, not for New Zealand. The coordinates transformation distorted and stretched the map.



Then I changed the entire map projection to CRS **EPSG:5132 Tokyo 1892** and this time the map is not distorted or stretched but the placement is misaligned and incorrect. The misalignment happened because of the coordinate systems. EPSG:4326 is the standard global latitude-longitude system based on Lat/long (Geodetic alias) method. Though the EPSG:5132 is also based on the same method but it is static and relies on a datum which is plate-fixed. That's why the layer projection on map was misaligned. This CRS is designed specifically for Tokyo so that it is not working correctly for New Zealand or any other geolocation.



Then for the third time I changed the whole map projection CRS to **ESGP:27700 – OSGB36 British National Grid**. As a result, the view of map was changed. Also this time the map is not stretched or distorted but the projection was totally misaligned due to the CRS. This is a static CRS based on transverse mercator method and it uses meters as unit. This CRS is designed for United Kingdom and doesn't work good on New Zealand or any other place.

